Module 1 Physics Notes: Ideal Magnetohydrodynamic Equations

Physics-Informed Neural Network for Two-Dimensional MHD Evolution

Purpose: derive the equations used by the conventional MATLAB solver before the PINN stage.

Scope of this note. This file derives the **ideal magnetohydrodynamic (MHD)** equations in the conservative form used by the two-dimensional finite-volume solver. The note emphasizes physical meaning, not just algebra. Every symbol used in the derivation is defined in the symbol table below.

1 Symbol and variable table

Symbol	Name	Meaning
ρ	mass density	Mass per unit volume of the plasma.
p	gas pressure	Thermal pressure of the fluid.
γ	ratio of specific heats	Closure parameter for an ideal gas, commonly $\gamma = 5/3$.
$\mathbf{u} = (u_x, u_y, u_z)$	fluid velocity	Plasma bulk-flow velocity.
$\mathbf{B} = (B_x, B_y, B_z)$	magnetic field	Magnetic-field vector.
E	total energy density	Sum of internal, kinetic, and magnetic energy densities.
p_{tot}	total pressure	Sum of gas pressure and magnetic pressure.
$\mathbf{U}$	conserved-state vector	Vector of conserved variables updated by the finite-volume solver.
$\mathbf{F}$	x-direction flux vector	Flux of conserved quantities through a face normal to the x-direction.
$\mathbf{G}$	y-direction flux vector	Flux of conserved quantities through a face normal to the y-direction.
u_x, u_y, u_z	velocity components	Components of bulk fluid velocity in the x, y, and z directions.
B_x, B_y, B_z	magnetic-field components	Components of magnetic field in the x, y, and z directions.
$\mathbf{u} \cdot \mathbf{B}$	velocity–field dot product	Measures how much the flow is aligned with the magnetic field.
$\nabla \cdot \mathbf{B}$	magnetic divergence	Must be zero in ideal MHD because magnetic monopoles are not observed.
x, y, t	coordinates and time	Two spatial coordinates and time variable used by the solver.

2 Why magnetohydrodynamics is the right model here

2.1 From particle motion to fluid fields

A plasma is a gas made of charged particles. In principle, one could follow every ion and electron individually, but that is usually far too expensive and also unnecessary when the physical scales of interest are much larger than microscopic particle orbits. Magnetohydrodynamics replaces the enormous collection of particles with smooth continuum fields such as density, velocity, pressure, and magnetic field.

The central physical assumption is that, on scales large compared with microscopic kinetic lengths, the plasma behaves like a conducting fluid. The fluid carries momentum and internal energy, while the magnetic field stores energy and exerts stress on the fluid.

Physical picture. Ordinary gas dynamics is a fluid pushed by pressure gradients and inertia. MHD is a fluid pushed by pressure gradients, inertia, and magnetic stresses. The magnetic field behaves partly like a pressure and partly like a tension-bearing elastic structure.

2.2 Why the equations are written in conservative form

The numerical method used in the project is a finite-volume method. A finite-volume method evolves averages of conserved quantities inside computational cells. For that reason, the governing equations are written in a conservative balance-law form: the time change inside a cell equals the net flux through the cell boundaries.

That is why the variables updated numerically are not u_x, u_y, u_z directly, but the conserved densities ρu_x, ρu_y, ρu_z, and the total energy density E.

3 The conserved-state vector

The solver uses the conserved variable vector

$$\mathbf{U} = \begin{bmatrix} \rho \\ \rho u_x \\ \rho u_y \\ \rho u_z \\ E \\ B_x \\ B_y \\ B_z \end{bmatrix}. \tag{1}$$

Each component of (1) has a direct physical meaning:

- ρ is the amount of mass per unit volume.
- ρu_x, ρu_y, and ρu_z are momentum densities.
- E is the total energy density.
- B_x, B_y, and B_z are stored as part of the state because the magnetic field evolves dynamically.

3.1 Why the magnetic field appears in the state vector

In MHD the magnetic field is not imposed from the outside as a fixed background. It evolves along with the plasma. That is why the induction equation must be solved together with the fluid equations. Numerically, that means the magnetic-field components must be advanced in time just like the density, momentum, and total energy.

4 Derivation of the total energy density

The total energy density in ideal MHD is

$$E = \frac{p}{\gamma - 1} + \frac{1}{2}\rho\left(u_x^2 + u_y^2 + u_z^2\right) + \frac{1}{2}\left(B_x^2 + B_y^2 + B_z^2\right). \tag{2}$$

We now derive each term.

4.1 Internal energy term

For an ideal gas, the internal energy density is

$$e_{\text{int}} = \frac{p}{\gamma - 1}. \tag{3}$$

This relation is the ideal-gas closure used by the solver. It converts pressure into internal thermal energy density.

4.2 Kinetic energy term

The kinetic energy per unit volume is mass density times kinetic energy per unit mass:

$$e_{\text{kin}} = \rho\left(\frac{1}{2}|\mathbf{u}|^2\right) = \frac{1}{2}\rho\left(u_x^2 + u_y^2 + u_z^2\right). \tag{4}$$

4.3 Magnetic energy term

In normalized ideal MHD units, the magnetic energy density is

$$e_{\text{mag}} = \frac{1}{2}|\mathbf{B}|^2 = \frac{1}{2}\left(B_x^2 + B_y^2 + B_z^2\right). \tag{5}$$

4.4 Why these three terms add

The plasma can store energy in three physically distinct ways:

1. thermal disorder of particles, represented by the internal energy,
2. bulk organized motion of the fluid, represented by kinetic energy,
3. the electromagnetic field, represented here by magnetic energy.

Ideal MHD treats these as interconvertible forms of energy. For example, a compressed magnetic field can accelerate the fluid, or kinetic motion can bend field lines and increase magnetic energy.

5 Derivation of the conservative equations

The ideal MHD equations used in the solver are

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}(\mathbf{U})}{\partial x} + \frac{\partial \mathbf{G}(\mathbf{U})}{\partial y} = 0. \tag{6}$$

This compact vector equation really contains eight coupled scalar equations. We now explain them one by one.

5.1 Mass conservation

Mass conservation states that matter is not created or destroyed inside the fluid element. In differential form,

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0. \tag{7}$$

In two-dimensional Cartesian form,

$$\frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x} + \frac{\partial}{\partial y} = 0. \tag{8}$$

The first component of the flux vectors must therefore be the mass fluxes ρu_x and ρu_y.

5.2 Momentum conservation and the role of magnetic stress

If there were no magnetic field, the momentum equation would be the Euler equation of compressible gas dynamics. MHD adds magnetic stress. The momentum balance can be written as

$$\frac{\partial}{\partial t} + \nabla \cdot \left[\rho \mathbf{u}\mathbf{u} + \left(p + \frac{1}{2}|\mathbf{B}|^2 \right) \mathbf{I} - \mathbf{B}\mathbf{B} \right] = 0, \tag{9}$$

where $\mathbf{I}$ is the identity tensor.

Equation (9) is extremely important because it shows that the magnetic field enters in two distinct ways:

1. through an isotropic magnetic-pressure term $\frac{1}{2}|\mathbf{B}|^2\mathbf{I}$,
2. through a directional tension term $-\mathbf{B}\mathbf{B}$.

Interpretation of magnetic stress. Magnetic pressure tries to push outward from regions of strong magnetic field. Magnetic tension tries to straighten bent field lines, much like tension in a stretched string. MHD dynamics arises from the competition between fluid inertia, gas pressure, magnetic pressure, and magnetic tension.

5.3 Energy conservation

The energy equation in conservative form is

$$\frac{\partial E}{\partial t} + \nabla \cdot \left[(E + p_{\text{tot}})\, \mathbf{u} - (\mathbf{u} \cdot \mathbf{B})\mathbf{B} \right] = 0, \tag{10}$$

where

$$p_{\text{tot}} = p + \frac{1}{2}|\mathbf{B}|^2. \tag{11}$$

The first term inside the energy flux, $(E + p_{\text{tot}})\mathbf{u}$, resembles the usual fluid-energy advection term but with total pressure instead of only gas pressure. The second term, $(\mathbf{u} \cdot \mathbf{B})\mathbf{B}$, accounts for the part of electromagnetic energy transport coupled to the direction of the field.

5.4 Induction equation

The induction equation for ideal MHD is

$$\frac{\partial \mathbf{B}}{\partial t} - \nabla \times (\mathbf{u} \times \mathbf{B}) = 0. \tag{12}$$

Using vector identities and the divergence-free condition $\nabla \cdot \mathbf{B} = 0$, this can be written in conservative component form. In the two-dimensional solver, the magnetic-field fluxes are represented directly in the $\mathbf{F}$ and $\mathbf{G}$ vectors.

5.5 Divergence-free constraint

An additional constraint must hold:

$$\nabla \cdot \mathbf{B} = 0. \tag{13}$$

In two dimensions this becomes

$$\frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} = 0. \tag{14}$$

This is not an optional detail. It expresses the empirical absence of magnetic monopoles. Numerically, if this constraint is violated too strongly, the solver can produce unphysical forces. That is why the project includes a diagnostic $\nabla \cdot \mathbf{B}$ monitor even before more advanced divergence-control methods are added.

6 Flux vectors used by the MATLAB code

Using the definitions above, the x-direction flux vector is

$$\mathbf{F} = \begin{bmatrix} \rho u_x \\ \rho u_x^2 + p_{\text{tot}} - B_x^2 \\ \rho u_x u_y - B_x B_y \\ \rho u_x u_z - B_x B_z \\ (E + p_{\text{tot}}) u_x - (\mathbf{u} \cdot \mathbf{B}) B_x \\ 0 \\ u_x B_y - u_y B_x \\ u_x B_z - u_z B_x \end{bmatrix}. \tag{15}$$

The y-direction flux vector is

$$\mathbf{G} = \begin{bmatrix} \rho u_y \\ \rho u_y u_x - B_y B_x \\ \rho u_y^2 + p_{\text{tot}} - B_y^2 \\ \rho u_y u_z - B_y B_z \\ (E + p_{\text{tot}}) u_y - (\mathbf{u} \cdot \mathbf{B}) B_y \\ u_y B_x - u_x B_y \\ 0 \\ u_y B_z - u_z B_y \end{bmatrix}. \tag{16}$$

6.1 Why the B_x flux in x is zero, and the B_y flux in y is zero

The MATLAB tests check that the sixth component of $\mathbf{F}$ is zero and the seventh component of $\mathbf{G}$ is zero. This is not arbitrary. Those zeros are built into the induction equation when written in conservative Cartesian form. They are a useful structural consistency check because a coding mistake in the magnetic flux terms often shows up immediately in these entries.

7 Primitive and conserved variables

The solver moves back and forth between two representations:

$$\text{primitive variables: } (\rho, u_x, u_y, u_z, p, B_x, B_y, B_z), \tag{17}$$

$$\text{conserved variables: } (\rho, \rho u_x, \rho u_y, \rho u_z, E, B_x, B_y, B_z). \tag{18}$$

7.1 Why primitive variables are useful

Primitive variables are the natural variables for physical interpretation and for initializing test problems. For example, it is far easier to specify an Alfvén-wave perturbation in terms of u_z and B_z than in terms of total energy density.

7.2 Why conserved variables are useful

Finite-volume methods update cell averages of conserved densities. That is why the numerical solver evolves the conserved state vector $\mathbf{U}$.

7.3 Recovering pressure from the conserved state

The pressure-recovery formula used by the MATLAB code comes directly from solving (2) for p:

$$p = (\gamma - 1)\left[E - \frac{1}{2}\rho\left(u_x^2 + u_y^2 + u_z^2\right) - \frac{1}{2}\left(B_x^2 + B_y^2 + B_z^2\right)\right]. \tag{19}$$

This is an important formula because any numerical error in total energy immediately affects the recovered pressure.

8 Why these equations are appropriate for the first solver stage

For the first solver stage, the goal is to verify wave propagation and then demonstrate a nonlinear magnetic-pressure pulse. Ideal MHD is the right starting point because it captures the key coupling between the plasma flow and the magnetic field while keeping the system manageable enough to implement and verify in MATLAB.

The following effects are intentionally excluded at this stage:

- resistivity,
- viscosity,
- thermal conduction,
- Hall terms,

- kinetic particle effects.

Those may matter in more advanced plasma models, but omitting them here is scientifically reasonable because the first objective is to construct a clean baseline solver whose behavior can be understood and verified.

9 Connection to the Alfvén-wave verification case

The first verification case in the project is a small-amplitude shear Alfvén wave. That test is ideal because it checks whether the code correctly transmits coupled perturbations in transverse velocity and magnetic field at the expected Alfvén speed. In other words, it directly tests whether the momentum equation, induction equation, and magnetic stresses have been implemented consistently.

Practical solver takeaway. The MATLAB files `primitive_to_conserved.m`, `conserved_to_primitive.m`, `compute_pressure.m`, `compute_total_energy.m`, `flux_x.m`, and `flux_y.m` are direct numerical realizations of the physics derived in this note. The solver is not just "doing numerics"; it is advancing the exact conservative balances derived above.

10 Summary

The core ideas of this note are:

1. MHD treats a conducting plasma as a continuum fluid coupled to a magnetic field.
2. The finite-volume solver evolves conserved quantities, so the governing equations must be written in conservative form.
3. The total energy density contains internal, kinetic, and magnetic contributions.
4. Magnetic fields affect momentum through both pressure-like and tension-like stresses.
5. The divergence-free condition $\nabla \cdot \mathbf{B} = 0$ is physically essential and numerically important.
6. The flux vectors used in the MATLAB code follow directly from the conservative ideal-MHD equations.